

3D Application for Animal Tracking

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Abstract

New visualization trends bring new light to the way environmental monitoring data is exploited today. The presented application is an enhancement of the user experience for the visualization and analysis of the Argos service for animal tracking. Argos is a satellite-based system that collects data from Platform Terminal Transmitters, and distributes sensor and location data in order to understand the distribution of wildlife in a given territory.

Virtual globes constitute a visualization platform of geospatial information projected onto the Earth Digital Elevation Model (DEM) in three dimensions, users interactively rotate, zoom and pan the available data, usually projected on a given background that provides a geographic context. The 3D platform allows access to animal tracking data. It also permits the extraction of specific information, path animation and data download from a dedicated and secure platform, with the objective of facilitating the extensive use of this source of information.

The animal tracking 3D desktop application has been built with the World Wind SDK, an open source virtual globe built in Java/OpenGL and developed by the National Aeronautics and Space Administration (NASA). The user can combine tracking data with layers from any public server implementing the WMS (Web Mapping Service) protocol of the OGC (Open Geospatial Consortium) through a tab menu able to handle the layers and opacity. Animal tracks can be animated individually alongside the elevation profile of the trajectory with the error ellipsoids of each position of the animal. As for the input/output capabilities of the tool, the CSV format is used as default in order to be compatible with existing operational services. KML is also available to enhance interoperability within GIS communities. The application allows the possibility to query directly from the WebServices that hold the animal data through a REST API using SOAP (Simple Object Access Protocol) with authentication.

The new 3D approach to the way data is visualized and the possibility to overlay a wide range of data from open WMS servers, allows users of the service to have a more realistic and integrated vision of their animal positioning data.